

$EO_T = KL$ projection of cost into feasible set



other types of projections

Iterative Bregman Projections

Repeat until convergence:

$$\pi^{(\ell+1)} \leftarrow \text{Proj}_{\mathcal{C}_a^1}^{\text{KL}}(\pi^{(\ell)}); \quad \pi^{(\ell+2)} \leftarrow \text{Proj}_{\mathcal{C}_b^2}^{\text{KL}}(\pi^{(\ell+1)})$$

Theorem [Bregman IP Convergence]:

$$\pi^{(\ell)} \rightarrow \pi^* = \arg \min_{\pi \in \mathcal{C}_{\mathbf{a}}^1 \cap \mathcal{C}_{\mathbf{b}}^2} \text{KL}(\pi \mid \mathbf{K})$$

Sinkhorn: Projection View

- Let $\mathcal{C}_a^1 = \{\pi \mid \pi \mathbf{1}_m = \mathbf{a}\}$, $\mathcal{C}_b^2 = \{\pi \mid \pi^\top \mathbf{1}_n = \mathbf{b}\}$. Then: $\Pi(\mathbf{a}, \mathbf{b}) = \mathcal{C}_a^1 \cap \mathcal{C}_b^2$

- Can solve via Bregman Iterative Projections

Iterative Bregman Projections

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- One can show: Sinkhorn iteration $\Leftrightarrow$ Bregman projections
- Therefore: **EOT = KL projection of cost into feasible set** (transport polytope)
- Typically, we want to use Sinkhorn since it operates on vectors
- But this view opens the door to **other types of projections** and constraint sets

Sinkhorn: Historical Perspective